

Poster Presentation Reference

Core Message

We train subject-specific encoders for EEG, MEG, and fMRI that map brain responses to the same 512-dimensional CLIP image space. Once each modality is aligned to this shared space, we can evaluate image retrieval within a modality and cross-modality conversion by directly comparing embeddings.

Key distinction: there is no separate neural translation network for modality conversion. Conversion works because each modality encoder learns to place responses to the same image near the same CLIP-aligned point.

Model Architecture By Modality

Modality	Input	Encoder	Pooling / projection	Output
EEG	THINGS-EEG2 trials, currently 63 channels x 100 timepoints.	CBraMod / Criss-Cross Transformer backbone, loaded from pretrained weights when available. Inputs are resampled to 200 timepoints and treated as one patch.	Four pseudo-anatomical channel regions are averaged, passed through region-specific linear layers, and combined with learned softmax region weights. Then Linear -> ReLU -> Linear projects to CLIP size.	L2-normalized 512-D CLIP-space brain embedding.
MEG	THINGS-MEG epochs, 271 sensors x time after crop/resample and channel-wise standardization.	Dedicated Temporal CNN, not CBraMod. First uses 1x1 sensor projection to 256 hidden channels, then depthwise temporal stem and four residual temporal blocks with dilations 1, 2, 4, and 8.	Learned attention over time pools the temporal features, then LayerNorm and Linear -> GELU -> Dropout -> Linear.	L2-normalized 512-D CLIP-space brain embedding.
fMRI	Single-trial ICA-beta voxel vectors. Top 15,000 highest-variance voxels are selected using train trials only, then z-scored with train statistics.	Bottleneck MLP: Linear(V, 2048) -> LayerNorm -> GELU -> Dropout, then a residual 2048-D MLP block.	Final Linear(2048, 512). Dropout is 0.5 in the fMRI model.	L2-normalized 512-D CLIP-space brain embedding.

Alignment Model

The shared alignment target is frozen CLIP ViT-B/32 image embeddings.

Component	What to remember
CLIP target	Each stimulus image is encoded once with CLIP ViT-B/32 and cached as a normalized 512-D vector.
Brain encoder	Each modality has its own subject-specific model that predicts a 512-D vector from brain data.
Shared space	Brain predictions and CLIP image vectors are both L2-normalized, so dot product equals cosine similarity.
Temperature	A learnable <code>logit_scale</code> starts from CLIP's standard temperature init, <code>tau = 0.07</code> .
Loss	Symmetric CLIP / InfoNCE loss: brain-to-image and image-to-brain cross entropy over the batch, with matching pairs on the diagonal.

Useful phrase: “The alignment model is the combination of a modality-specific brain encoder, a shared CLIP target space, unit-normalized embeddings, and a symmetric contrastive loss.”

Evaluation Terms

Retrieval

Retrieval asks whether a brain embedding can identify the image that caused it.

- Repeated trials for the same `image_id` are averaged at evaluation time.
- The averaged brain embedding is compared against all candidate CLIP image embeddings.
- Top-1 means the correct image is ranked first.
- Top-5 means the correct image is anywhere in the five nearest candidates.
- In these tables, “class” means exact `image_id`, not semantic concept.

CLIP2 / CLIP 2-Way

CLIP2 means CLIP 2-way: a pairwise ranking metric.

- For each true brain-image pair, compare the true similarity against every mismatched candidate similarity.
- Score the fraction of pairwise comparisons where the true match wins.
- Average across all images.
- Chance is about 50%, not 1 / candidate count, because it is a two-choice comparison repeated over many distractors.
- This is useful when 200-way Top-1 is harsh but the correct item is still generally ranked above distractors.

CLIP2 Normalized / CLIP 2-Way Normalized

Normalized CLIP 2-way asks how much of the target modality's own decodability is preserved during conversion.

- **2-way / matched retrieval**: conversion CLIP 2-way divided by the target modality's retrieval CLIP 2-way on the same shared pool.
- **2-way / native retrieval**: conversion CLIP 2-way divided by the target modality's full native retrieval CLIP 2-way.
- Values near 1 mean conversion is close to the target modality's own image-decoding ceiling under that reference.

- Matched retrieval is the fairest comparison for shared-pool conversion because the shared-pool models can be weaker than full native models.

2-Way Conversion

Conversion evaluates whether one modality’s embedding can retrieve the matching embedding from another modality.

- Example: EEG -> MEG compares each EEG test embedding to all MEG test embeddings for the same shared images.
- The correct match is the MEG embedding with the same `image_id`.
- Forward and reverse directions are both reported because the similarity matrix can be read either way.
- This is direct embedding-space comparison, not reconstruction of raw MEG, fMRI, or EEG signals.

Training Protocol

Item	What we did
CLIP cache	Built a universal CLIP ViT-B/32 cache over the THINGS image manifests. CLIP features are frozen targets.
Subject training	Trained separate checkpoints for each subject and modality.
Objective	Symmetric contrastive loss between predicted brain embedding and cached CLIP image embedding.
Optimizer	AdamW, learning rate 3e-4, weight decay 1e-3.
Epochs	60 epochs per modality by default.
Batch sizes	EEG 32, MEG 32, fMRI 256.
End-to-end training	EEG fine-tunes the pretrained CBraMod path end-to-end; MEG and fMRI train their modality-specific alignment models end-to-end.
Checkpoint selection	Best checkpoint selected by validation MRR for the current config.
Full retrieval models	Trained on each modality’s native train/val split and evaluated on native test images.
Conversion models	Trained separately with <code>--shared-only</code> on manifest-backed shared conversion pools.
Conversion split	Concept holdout with 100 validation concepts and 200 test concepts; evaluation is exact-image retrieval.

Data Splits To Remember

Benchmark	Candidate set	Important detail
EEG full retrieval	200-way	Official THINGS-EEG2 train/test release. Test images and concepts are novel.
MEG full retrieval	200-way	Fixed EEG-style held-out image split. Test images and concepts are novel.
fMRI full retrieval	100-way	Official THINGS-fMRI train/test split. Test images are novel, but concepts appear in train.

Benchmark	Candidate set	Important detail
Pairwise conversion	200-way	Shared image pools are built from images train-eligible in both modalities.
3-way conversion	200-way	Stricter shared pool across EEG, MEG, and fMRI.

Conversion pool sizes:

Pool	Trainable pool	Train split	Val split	Test split
EEG-MEG	16,540 images / 1,654 concepts	13,540 images / 1,354 concepts	100 images / 100 concepts	200 images / 200 concepts
EEG-fMRI	6,330 images / 633 concepts	3,330 images / 333 concepts	100 images / 100 concepts	200 images / 200 concepts
MEG-fMRI	8,640 images / 720 concepts	5,040 images / 420 concepts	100 images / 100 concepts	200 images / 200 concepts
EEG-MEG-fMRI	6,330 images / 633 concepts	3,330 images / 333 concepts	100 images / 100 concepts	200 images / 200 concepts

Main Results To Remember

Native retrieval:

Modality	Candidate set	Top-1	Top-5	CLIP 2-way
EEG	200-way	12.95 +/- 3.29	39.35 +/- 7.70	90.57 +/- 2.12
MEG	200-way	8.00 +/- 4.08	25.12 +/- 11.04	82.95 +/- 5.24
fMRI	100-way	33.33 +/- 6.60	67.67 +/- 5.44	93.87 +/- 1.43

Pairwise conversion:

Shared pool	Direction	Top-1	Top-5	CLIP 2-way	2-way / matched retrieval
EEG-fMRI	EEG -> fMRI	1.52 +/- 1.10	5.93 +/- 2.05	61.69 +/- 4.79	0.868 +/- 0.068
EEG-fMRI	fMRI -> EEG	1.27 +/- 0.92	5.95 +/- 2.52	61.58 +/- 4.92	0.901 +/- 0.072
EEG-MEG	EEG -> MEG	2.92 +/- 1.49	12.72 +/- 4.33	72.89 +/- 5.96	0.924 +/- 0.030
EEG-MEG	MEG -> EEG	3.19 +/- 1.73	12.60 +/- 4.91	73.41 +/- 5.77	0.842 +/- 0.064
MEG-fMRI	MEG -> fMRI	1.00 +/- 0.74	5.17 +/- 2.55	60.02 +/- 5.52	0.807 +/- 0.077
MEG-fMRI	fMRI -> MEG	1.21 +/- 0.90	6.29 +/- 3.31	60.27 +/- 5.77	0.925 +/- 0.045

Takeaways

- All three modalities decode visual identity above chance in CLIP space.
- fMRI is strongest for native retrieval, likely because voxel responses carry high spatial specificity for visual content.
- EEG and MEG have lower Top-1 than fMRI, but their CLIP 2-way scores are high, showing that correct images are usually ranked above many distractors.
- EEG-MEG conversion is the strongest pairwise result, which makes sense because both are fast electrophysiological modalities.

- EEG-fMRI and MEG-fMRI conversion are harder, but still above chance in 2-way terms.
- Low conversion Top-1 is not the whole story: 200-way chance is 0.5%, and normalized CLIP 2-way shows conversion often preserves a large fraction of matched-pool decodability.
- The 3-way shared pool is stricter, so raw conversion scores drop, but it is the cleanest protocol for comparing all modalities on the same images.

Why This Research Matters

- It tests whether different brain measurement technologies contain a shared visual representational structure.
- It creates a common language for EEG, MEG, and fMRI by mapping all of them into CLIP’s semantic image space.
- It helps bridge modalities with different strengths: EEG/MEG have high temporal resolution, while fMRI has stronger spatial specificity.
- It can support cross-modal transfer, missing-modality prediction, and multimodal brain decoding without requiring a new raw-signal translator for every pair.
- It points toward reusable neural representation models that can compare subjects, datasets, and recording technologies in a shared embedding space.
- Longer term, this kind of alignment can contribute to brain-computer interfaces, stimulus decoding, cognitive neuroscience benchmarks, and multimodal neuroimaging tools.

Quick Answers For Poster Questions

Question	Short answer
What is being aligned?	Brain responses from EEG, MEG, and fMRI are aligned to frozen CLIP image embeddings.
Why CLIP?	CLIP gives a strong semantic visual space, so image similarity is meaningful beyond pixel matching.
Is conversion a separate model?	No. Conversion is direct retrieval between modality embeddings after each encoder is trained into the shared CLIP space.
Why use CLIP 2-way?	It captures pairwise ranking quality and is less brittle than 200-way Top-1. Chance is about 50%.
Why normalized 2-way?	It tells whether conversion is close to the target modality’s own retrieval performance on the same benchmark.
What is the main result?	Native decoding works for all modalities, and cross-modal conversion preserves meaningful image information, especially EEG-MEG.
Biggest limitation?	Conversion depends on shared image pools and subject-specific training; raw Top-1 remains low for harder cross-modal pairs.
Next step?	Stronger shared encoders, more subjects, better multimodal training objectives, and tests on unseen concepts or larger candidate sets.